

ASJAD KHAN

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SUMMARY

Data Engineer with 8+ years of experience building real-time and batch data platforms at scale.

Expert in Kafka, Spark, Azure Databricks, and Delta Lake, with a proven track record of improving pipeline performance and system reliability. Strong in cloud-native architecture (AKS, Terraform, Azure networking) and high-throughput ETL design. Leverages AI-driven tools and automation to accelerate development and optimize data workflows. Designed and deployed end-to-end Azure data platforms integrating real-time streaming, batch processing, and cloud-native infrastructure.

SKILLS

Big Data & Processing: Apache Kafka, Apache Spark, Databricks, Delta Lake, ETL Pipelines

Cloud Platforms: Microsoft Azure (AKS, Data Factory, Data Lake, Azure SQL, Storage), GCP

AI & Automation: AI-assisted development, prompt engineering, intelligent pipeline optimization

Orchestration: Apache Airflow, Oozie

Infrastructure & DevOps: Terraform, Jenkins, GitHub Actions, Maven, Git

Networking & Security: Azure VNet, Subnets, NSGs, Firewalls, Private Endpoints, Private Link Service

Databases: MongoDB, Hive, HBase, MySQL, Oracle

Languages: Java, Python, SQL

Frameworks: Spring Boot, Spring Batch, Hibernate

EXPERIENCE

Data Engineer

Optum (UnitedHealth Group) — Chicago, IL.

Nov 2021 – Present

- Owned and scaled real-time data pipelines (Kafka + Spark) processing millions of records daily across distributed systems.
- Architected and implemented Databricks + Delta Lake platform for scalable batch and streaming workloads with ACID-compliant processing.
- Deployed and managed containerized applications on AKS, improving scalability and availability of production systems.
- Designed and automated Azure infrastructure using Terraform, including VNet, NSGs, Firewalls, and Private Endpoints
- Owned orchestration layer using Apache Airflow, enabling reliable scheduling, monitoring, and recovery of ETL workflows.
- Integrated MongoDB for low-latency, high-throughput data access in distributed systems.
- Implemented CI/CD pipelines, reducing deployment time and increasing release reliability.
- Leveraged AI-assisted development to automate pipeline creation, reduce development time, and improve engineering productivity
- Developed Python-based data processing and automation scripts to support ETL workflows and pipeline orchestration.
- Wrote and optimized complex SQL queries, including joins, aggregations, and window functions, for efficient large-scale data processing.
- Designed and implemented data models (fact/dimension tables) to support scalable analytics and reporting use cases.

PROJECTS

United Health Group (Optum: RQNS Platform)

- Architected and deployed end-to-end Azure data platform supporting real-time and batch workloads using Kafka, Spark, and Databricks.
- Designed secure network architecture (VNet, Subnets, NSGs, Application Gateway) for enterprise-grade deployments.
- Designed and implemented Airflow-based orchestration framework to manage complex multi-system data workflows
- Applied AI-driven techniques for pipeline optimization, anomaly detection, and intelligent monitoring

United Health Group (Optum:Member Micro-services)

- Designed event-driven architecture using Kafka to enable scalable and reliable data streaming across distributed systems
- Engineered high-performance Spark jobs in Java for large-scale data transformation
- Designed low-latency data processing systems handling diverse, high-volume datasets.
- Built secure REST APIs with OAuth2, supporting scalable microservices architecture.

United Health Group (Optum:MGM Platform)

- Designed Spring Batch pipelines, reducing batch processing time by 25%.
- Optimized application performance using multithreading and Java 8 features.

Java Software Engineer

Cognizant Technology Solutions

July 2016 – Aug 2019

- Delivered enterprise-grade backend systems for insurance clients with high availability and performance requirements
- Developed scalable backend systems using Java, Spring Boot, and Hibernate.
- Built REST APIs and contributed across full SDLC lifecycle.
- Optimized database queries and improved system performance.

EDUCATION

Master of Computer Science, California State University San Marcos

(GPA:3.8) 2019-2021

Bachelors in Technology, SRM University Chennai

(GPA:3.5) 2012-2016

LEADERSHIP

- Led data engineering initiatives using Kafka and Spark, achieving 40% improvement in data flow efficiency, 30% increase in processing performance, and 25% reduction in batch processing time.
- Architected and deployed end-to-end Azure infrastructure, including AKS, VNet, Subnets, NSGs, Firewalls, and Application Gateway, along with Terraform-based automation and Airflow pipelines for scalable, secure, and reliable data workflows.
- Drove AI integration within the team, enabling AI-assisted data pipeline optimization, intelligent monitoring, and productivity improvements, while managing full SDLC to reduce development time by 15% and improve system reliability by 20%.